Aayaann Kausar

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Education

Indian Institute of Technology (IIT) Kharagpur — Kharagpur, India
B.Tech in Chemical Engineering — *2024 - Present*

Sri Chaitanya Techno School
Central Board of Secondary Education (Class 12th) — *2022 - 2024*

Awards and Achievements

JEE Main 2024 — Top **0.29%** rank among 1.4M+ candidates

JEE Advanced 2024 — Top **3.2%** rank among 250K+ candidates

WBJEE 2024 — Top **0.20%** rank among 113K+ candidates

Projects

VoRTeX: Production AI Execution Engine | GitHub — July 2026 – August 2026
- Architected **distributed AI execution engine** using **Event Sourcing**, **CQRS**, and Dynamic DAG orchestration with **0.015 ms** projection latency
- Built multi-provider **LLM Gateway** with **semantic caching**, **KV-cache affinity**, **Redis TTL leases**, and circuit-breaker failover across OpenAI, Anthropic, Gemini, and local models
- Implemented **AI guardrails**, PII scrubbing, multi-tenant isolation, and **OpenTelemetry observability**, achieving **99.8%** prompt-injection precision and **91.5%** faithfulness agreement

Autonomous AI Co-Scientist | GitHub — May 2026 – June 2026
- Architected a LangGraph **multi-agent system** with async orchestration and Corrective RAG for scientific reasoning
- Designed **Elo tournaments** and meta-review loops, achieving **4.5/5** Novelty and **71.9** Scientific Hypothesis Index
- Benchmarked **10** hypotheses in 12 minutes, achieving **4.0/5** Therapeutic Impact via **105** LLM calls

Distributed In-Memory Key-Value Store | GitHub — May 2026 – June 2026
- Engineered **Redis-compatible C++20 server** over raw **TCP/RESP** with **sharded storage**, **TTL expiry**, and **O(1) approximate LRU eviction**
- Implemented **AOF persistence** and **primary-replica streaming replication** with **PSYNC-style offsets**, enabling deterministic **crash recovery**
- Built **INFO-based telemetry** and load benchmarking; validated **12,000 req/s**, **0.73 ms P50**, **2.11 ms P99**, and **242 ms rebuild**

Competition / Conference

HFT Market State Forecasting Challenge | *Wunder Fund* — Sep 2025 – Nov 2025
- Secured **9th global rank**, achieving **0.3920** R^2 on private leaderboard via uncertainty-weighted regression
- Designed **4-layer Transformer** with **MAE pretraining** and statistical context fusion for market state forecasting
- Accelerated inference via **torch.compile** and **Numba**, optimizing execution latency for high-frequency trading environments

Skills and Expertise

Languages: Python, C++, SQL
AI Engineering: LLMs, Agentic AI, Multi-Agent Systems, RAG, AI Evaluation
Distributed Systems: Event Sourcing, CQRS, Workflow Orchestration, Distributed Caching, Replication, Async Programming
Backend & Infrastructure: FastAPI, REST APIs, TCP/IP, Redis, PostgreSQL, Docker, Linux, OpenTelemetry
Machine Learning: PyTorch, Transformers, Scikit-learn, LightGBM, ONNX Runtime
Developer Tools: Git, GitHub Actions, LangGraph, LangChain, LangSmith

Coursework Information

Mathematics: Probability and Statistics, Linear Algebra, Advanced Calculus
Computer Science: Programming and Data Structures
Artificial Intelligence: Machine Learning, Deep Learning